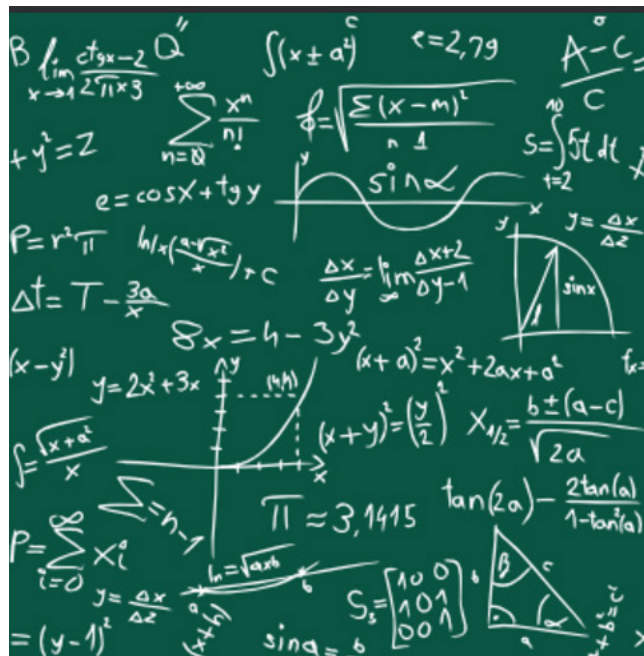


Exercises: maxima, minima, suprema and infima



- **Authors:** Fabio Furini and Gianluca Priori.
- **Book:** "Esercitazioni Analisi matematica 1", Marco Bramanti, Esculapio.

Contents

Exercise 1

Determine whether the set

$$A = [0, \sqrt{2}] \cap \mathbb{Q}$$

has a maximum, a minimum, a supremum, an infimum and, if so, determine these elements.

The minimum of A is 0. A has no maximum; its supremum is $\sqrt{2}$.

Exercise 2

Determine whether the set

$$A = [\sqrt{2}, \sqrt{3}] \cap \mathbb{Q}$$

has a maximum, a minimum, a supremum, an infimum and, if so, determine these elements.

A has neither a minimum nor a maximum. The infimum is $\sqrt{2}$, the supremum is $\sqrt{3}$.

Exercise 3

Determine whether the set

$$A = [0, \sqrt{2}] \cap (\mathbb{R} - \mathbb{Q})$$

has a maximum, a minimum, a supremum, an infimum and, if so, determine these elements.

A has no minimum; the infimum is 0. The maximum of A is $\sqrt{2}$.

Exercise 4

In each of the following cases, say whether the set

$$A = (-\infty, \sqrt{2}] \cap \mathbb{Q}$$

has a maximum, a minimum, a supremum, an infimum and, if so, determine these elements.

A is not bounded below, hence it has no infimum in $\mathbb{R}$. A has no maximum; the supremum is $\sqrt{2}$.

Exercise 5

Say whether the following set $A \subset \mathbb{R}$ has a maximum, a minimum, a supremum, an infimum, and whether it is bounded:

$$A = \left\{ \frac{2n+1}{n} : n \in \mathbb{N} \setminus \{0\} \right\}$$

From $(2n+1)/n = 2 + (1/n)$ we have that the maximum of A is attained for $n = 1$ and is equal to 3. A has no minimum; the infimum is 2. In particular, A is bounded.

Exercise 6

Say whether the following set $A \subset \mathbb{R}$ has a maximum, a minimum, a supremum, an infimum, and whether it is bounded:

$$A = \left\{ \frac{n-1}{n} : n \in \mathbb{N} \setminus \{0\} \right\}$$

From $(n-1)/n = 1 - (1/n)$ we have that the minimum of A is attained for $n = 1$ and is equal to 0. A has no maximum; the supremum is 1. In particular, A is bounded.

Exercise 7

Say whether the following set $A \subset \mathbb{R}$ has a maximum, a minimum, a supremum, an infimum, and whether it is bounded:

$$A = \left\{ \frac{3n^2+1}{n^2} : n \in \mathbb{N} \setminus \{0\} \right\}$$

From $(3n^2+1)/n^2 = 3 + (1/n^2)$ we have that the maximum of A is attained for $n = 1$ and is equal to 4. A has no minimum; the infimum is 3. In particular, A is bounded.

Exercise 8

Say whether the following set $A \subset \mathbb{R}$ has a maximum, a minimum, a supremum, an infimum, and whether it is bounded:

$$A = \left\{ \frac{1}{n^2 + 1} : n \in \mathbb{N} \setminus \{0\} \right\}$$

The maximum value of A is attained for $n = 1$ and is equal to $1/2$. A has no minimum; the infimum is 0. In particular, A is bounded.

Exercise 9

Let

$$A = \left\{ \left| \frac{3x}{x+1} \right| : -\frac{1}{2} < x \leq 2 \right\}.$$

Choose the correct statements among the following:

$$(a) \sup A \notin A \qquad (b) \inf A = \min A = 2$$

$$(c) \max A = 3 \qquad (d) \text{ none of the other answers is correct}$$

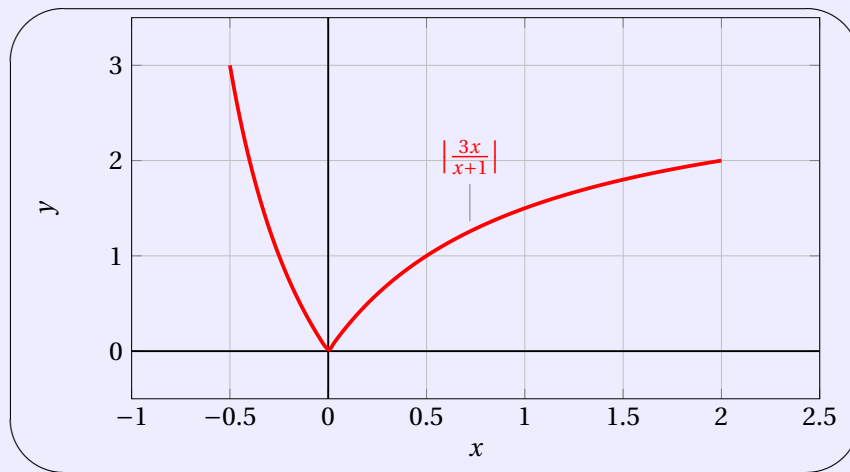
A is the set of values of $|f(x)|$ on the interval $(-1/2, 2]$, with $f(x) = \frac{3x}{x+1}$. From

$$\frac{3x}{x+1} = 3 - \frac{3}{x+1}$$

we have that the function f is strictly increasing on the given interval $(-1/2, 2]$, negative on $(-1/2, 0)$, zero for $x = 0$, and positive on $(0, 2]$. It follows that

$$|f(x)| = \left| \frac{3x}{x+1} \right|$$

is strictly decreasing on $(-1/2, 0]$, where it takes all the values in $[0, 3)$, and strictly increasing on $[0, 2]$, where it takes all the values in $[0, 2]$. The minimum value is therefore attained for $x = 0$ and is equal to 0. The supremum is 3; there is no maximum value. The correct answer is (a).



Exercise 10

For $I = [1/3, +\infty)$ consider the function

$$f : I \rightarrow \mathbb{R}, f(x) = \exp\left(\left|\frac{2x-1}{x}\right|\right).$$

Determine, among the following intervals, the set $J = f(I)$ of the values taken by f .

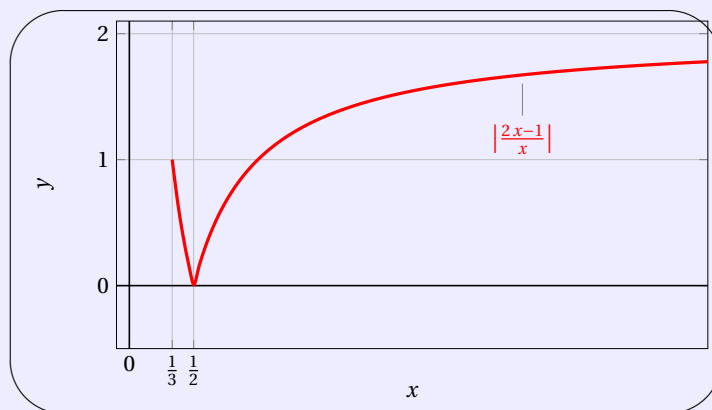
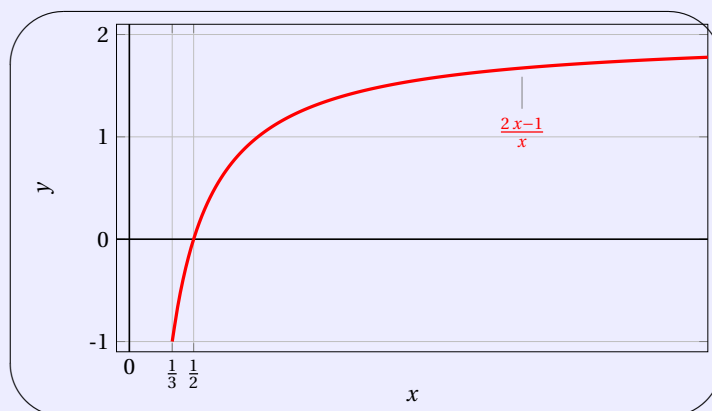
- | | | |
|--------------------|-------------------|----------------------|
| (a) $[e, e^2)$ | (b) $(e^{-2}, e]$ | (c) $[1, e^2)$ |
| (d) $(e, +\infty)$ | (e) $(0, e^2)$ | (f) another interval |

The function f has the same monotonicity behavior as

$$|g(x)| = \left| \frac{2x-1}{x} \right|$$

with

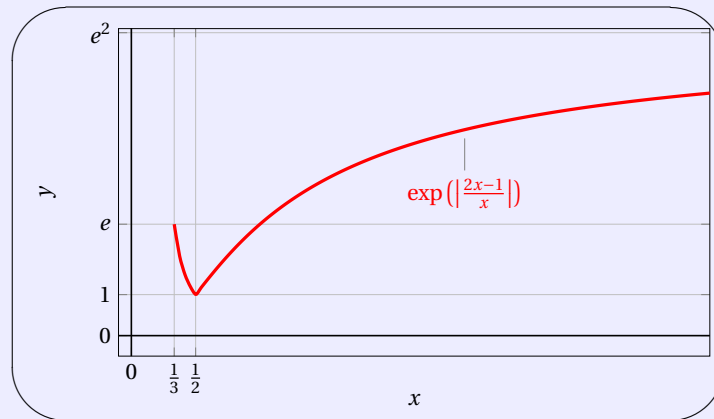
$$g(x) = \frac{2x-1}{x} = 2 - \frac{1}{x}.$$



The function g is strictly increasing on the interval $I = [1/3, +\infty)$, negative on $[1/3, 1/2)$, zero for $x = 1/2$, and positive on $(1/2, +\infty)$. It follows that $|g(x)|$ is strictly decreasing on $[1/3, 1/2]$, where it takes all the values in $[0, 1]$, and strictly increasing on $[1/2, +\infty)$, where it takes all the values in $[0, 2)$. The set of values $|g|(I)$ is $[0, 2)$, hence for $f(x) = e^{|g(x)|}$ we have

$$f(I) = [1, e^2).$$

The correct answer is (c).



More precisely, f is strictly decreasing on $[1/3, 1/2]$, where it takes all the values in $[1, e]$, and strictly increasing on $[1/2, +\infty)$, where it takes all the values in $[1, e^2)$.

Exercise 11

Let $A = A_+ \cup A_-$ with

$$A_+ = \left\{ x + \frac{2}{x} : x > 0 \right\}, \quad A_- = \left\{ x + \frac{2}{x} : x < 0 \right\}.$$

Choose the correct statements among the following:

- (a) $\sup A = +\infty$ (b) $\inf A_+ = 2\sqrt{2}$ (c) A_+ has no minimum
 (d) A_- has no maximum (e) none of the other answers is correct

Clearly, A_+ is not bounded above, hence statement (a) is correct (and statement (e) is false). For $x \neq 0$, we consider the equation

$$x + \frac{2}{x} = y,$$

$$x^2 - yx + 2 = 0.$$

There are solutions $x \in \mathbb{R}$ if and only if $y \in (-\infty, -2\sqrt{2}] \cup [2\sqrt{2}, +\infty)$. Moreover, for every $y \in (-\infty, -2\sqrt{2}]$ the solutions x are negative, while for every $y \in [2\sqrt{2}, +\infty)$ the solutions x are positive. This means

$$A_+ = [2\sqrt{2}, +\infty), \quad A_- = (-\infty, -2\sqrt{2}].$$

Therefore (b) is true, (c) is false, (d) is false.

